Shepherd — 1-Page Judge Card

Pitch: Shepherd is a live evacuation system for Edmonton that routes people to safe shelters on real roads as a wildfire or tornado spreads with the wind — and keeps re-routing them when roads jam or become unsafe.

Vision: Static "leave west" fails when fire, wind, and traffic change. Commanders get a live map; residents get a phone route that updates mid-trip — hazard grows with wind, roads close early, demand is estimated, phones reroute away from fire and gridlock.

Tools & why

FastAPI	JSON API; in-memory (no DB/auth)
React + Vite	Dashboard :5173 + phone :5174
MapLibre + OSM	Free maps; GeoJSON [lon, lat]
OSMnx / NetworkX	Real Edmonton drive graph
SciPy / NumPy / Shapely	Dijkstra, BPR, fire geometry
OpenWeather + wind sim	Spread + forecast outlines
Cloudflare (`./demo.sh`)	QR join for judges

How it works

1. **Hazard:** wildfire/tornado + wind; sim `T+N`; forecast (T+60).
2. **Road rule:** close at `arrival_min − 30`. Routes: **ok** (60+ min) / **tight** / **blocked**.
3. **Phone:** snap to OSM → open shelter → poll ∼2s → `shouldReroute` mid-trip.
4. **Traffic:** BPR delay; anti-gridlock. Up to **100k cars** via projection (∼2.5k agents × weight). Demand = pin + zone density + day/time.
5. **Commander:** closed roads, traffic ribbons, live blue phone routes, cut-off banner.

Demo: 1× wildfire → pin near phone → Play → fire/closes/reroute toast → cut-off banner. Reset clears traffic. Edmonton ≈ −113.5, 53.5.

Curveballs

Production / life-safety?	Prototype, not certified. Clear margin + real OSM; needs agency validation, auth, persistence.
Where do 100k cars come from?	Zone density × footprint × daypart → people → cars; projected onto fewer agents for realtime.
Why not Google/Waze?	Hazard-timed closures, shelter safety, commander + public state in one loop.
Why partial colouring?	Only actionable layers: fire, closed, jam, active routes.
Flood / other?	Flood deferred; same architecture extends.
Fire model accuracy?	Wind-driven demo/forecast, not firefighting product; routing uses `arrival_min`.
Privacy?	No auth; temporary pins; priority = type + last 4.
What's novel?	Wind hazard → timed closures → live phone reroute → commander cut-offs on real Edmonton OSM.
Next?	Census demand, websockets, agency roles, live shelter intake.

If you freeze: restate pitch → point fire / closed roads / blue route → one reroute → "happy to go deeper on routing, traffic, or demand."